Towards Efficient Smart Grid Operations: An Ensemble AI Model for Energy Forecasting

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Abstract: Accurate forecasting of energy consumption in smart grid environments is vital for efficient energy management and sustainable resource allocation. Addressing the challenge of modeling complex temporal and nonlinear patterns in energy usage, this research proposes a hybrid ensemble framework combining deep learning and gradient boosting techniques. Utilizing the Smart Grid Monitoring Dataset from Kaggle, the study incorporates rigorous preprocessing, feature engineering, and model optimization. Time-series features were extracted and normalized, with models such as Long Short-Term Memory (LSTM), CNN-GRU hybrids, and Light Gradient Boosting Machine (LightGBM) individually evaluated. The proposed ensemble model integrates GRU and LightGBM predictions using a weighted combination, capturing both temporal dependencies and structured feature patterns. Experimental results demonstrate that the ensemble model outperforms individual models, achieving an R2 score of 0.95, MAE of 0.12, and RMSE of 0.050. These findings validate the effectiveness of combining sequence models with decision-tree-based learners for improved accuracy and robustness in energy forecasting. The study contributes a scalable, data-driven solution for real-time energy prediction in smart grids.

SCIENCE AND TECHNOLOGY PUBLICATIONS

1 INTRODUCTION

In today's rapidly evolving energy landscape, smart grids have emerged as a transformative solution to meet the growing demand for efficient, reliable, and sustainable energy distribution. These intelligent systems integrate advanced metering infrastructure, sensor technologies, and data analytics to monitor and control energy flow in real time. A key challenge in smart grid management is accurately predicting energy consumption, which plays a critical role in optimizing power generation, minimizing energy waste, and ensuring system stability. With increasing global emphasis on energy conservation and smart infrastructure, the ability to forecast energy demand using artificial intelligence (AI) techniques has become not only desirable but essential.

Recent advancements in AI and machine learning have opened up new avenues for energy forecasting by enabling the modeling of complex, nonlinear, and time-dependent consumption patterns. Traditional statistical methods, while useful, often fall short when dealing with high-frequency and dynamic energy data common in smart grids. As a result, deep learning models such as Long Short-Term Memory (LSTM) networks, Convolutional Neural Networks (CNNs), and Gated Recurrent Units (GRUs) have gained traction for their superior performance in time-series prediction tasks. Additionally, ensemble learning methods like Light Gradient Boosting Ma- chine (LightGBM) have demonstrated exceptional capabilities in handling structured datasets and uncovering subtle patterns through gradient optimization.

Despite the progress made, existing approaches often struggle to balance predictive accuracy with computational efficiency. While deep learning models excel at capturing sequential dependencies, they may lack interpretability and require significant training time. Conversely, tree-based models offer fast and interpretable solutions but may fail to fully exploit temporal trends. This trade-off highlights a gap in current research - a need for hybrid frameworks that combine the strengths of various AI

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paradigms to deliver both precision and robustness. Addressing this gap is crucial for practical applications where prediction reliability directly impacts energy planning and distribution.

This study proposes a hybrid ensemble model that integrates GRU-based deep learning with the LightGBM algorithm to enhance energy consumption forecasting in smart grids. The research aims to assess the effectiveness of this ensemble approach in improving prediction accuracy over individual models. The central research question explored in this work is: Can a GRU-LightGBM ensemble model outperform standalone deep learning and boosting models in forecasting smart grid energy consumption? The hypothesis is that fusing temporal sequence learning with decision-tree-based gradient boosting can leverage complementary strengths for superior performance. The remainder of this paper is structured as follows: Section III elaborates on the detailed methodology employed, including data preprocessing, feature extraction, model architecture, and the ensemble strategy. Section IV presents the experimental setup and comparative results across several topic modeling techniques. Finally, Section V discusses the implications of our findings, followed by conclusions and directions.

2 RELATED WORKS

Cheng et al. proposed PowerNet, a neural architecture that integrates weather, calendar, and historical data to forecast power demand in smart grids. Their model outperformed traditional methods, reducing prediction error by up to 33.3% compared to Gradient Boosting Trees. Miah et al. developed a deep learning framework for short-term energy demand forecasting, integrating renewable energy sources. Their model achieved a mean absolute error (MAE) of just 1.4%, demonstrating high reliability and accuracy. Roy et al. implemented a Long Short-Term Memory (LSTM) network for energy demand prediction. Using four years of hourly data, their model achieved a mean absolute percentage error (MAPE) of 1.22%, outperforming ARIMA and linear regression baselines. Aguiar-Pe´rez and Pe´rez-Jua´rez reviewed deep learning techniques for demand forecasting in smart grids, covering CNNs, RNNs, and hybrid models. They concluded that deep learning consistently outperforms traditional statistical models in terms of RMSE and adaptability. Sarkar presented a hybrid deep learning model using CNN-LSTM for renewable load forecasting. Their method im- proved forecasting accuracy by 19% compared to conventional machine learning models. Kaur et al. surveyed state-of-the-art energy forecasting techniques in smart grids, highlighting the advantages of hybrid AI approaches. Their review noted that combining feature engineering with deep learning yields up to 20% better performance. Liu et al. introduced an XGBoost-based system for energy forecasting in renewable energy communities. Their method reduced forecasting errors by 22% and allowed better load scheduling to minimize energy costs. Zhang et al. proposed an attention-based deep learning model enhanced with federated learning and explainable AI (XAI). Their framework achieved up to 95% forecasting ac- curacy while maintaining privacy in distributed environments. Aqeel et al. implemented a hybrid LSTM-GRU model for residential load forecasting. The model showed high predictive power with a 97% accuracy and low training time due to efficient parallel computation. Moghaddam et al. compared several AI techniques for residential energy load forecasting and found that deep belief networks achieved the best accuracy, with a prediction error below 3%. Wang et al. explored the integration of AI with edge computing for real-time load prediction. Their edge-based AI model achieved a latency reduction of 30% and forecasting precision of 94.6%. Hossain et al. introduced a novel transformer-based approach for short-term electricity demand forecasting. The model outperformed LSTM and GRU baselines, reducing MAE by approximately 18.5%. Kamal et al. proposed a deep neural network model for long-term energy consumption prediction. Their model attained an R^2 score of 0.97, indicating high correlation between predicted and actual values. Qureshi et al. developed a stacked autoencoder model for energy demand prediction. Their results showed a significant reduction in RMSE when compared with single-layer networks and SVR. Alonso et al. built an ensemble deep learning model combining LSTM, CNN, and attention layers. The model achieved 96% accuracy on smart meter data and was robust across varying temporal resolutions.

3 METHODOLOGIES

The methodology adopted for this research encompasses a systematic approach involving data acquisition, preprocessing, feature extraction, model development, ensemble integration, and final evaluation. This structured workflow ensures a robust predictive framework for accurately forecasting energy consumption patterns within a smart grid environment.

3.1 Data Collection

The dataset employed in this study is the Smart Grid Monitoring Dataset, publicly available on Kaggle. This dataset simulates real-world operational data from a smart grid system and encapsulates essential electrical parameters such as voltage (V), current (I), frequency (f), power usage (P), and fault indicators over temporal intervals. With a compact size of approximately 193 KB and presented in CSV format, the dataset is both manageable and rich in information. Its design supports advanced machine learning applications, particularly time-series forecasting and anomaly detection. The relevance of this dataset stems from its ability to emulate the dynamics of modern smart grids, where energy demand and supply fluctuate continuously in response to various operational conditions.

3.2 Data Preprocessing

Data preprocessing plays a pivotal role in enhancing model accuracy and stability. The raw dataset, though well-structured, contained certain inconsistencies such as missing values and improperly formatted time stamps. Initially, any missing numerical entries were imputed using a rolling mean strategy over a fixed temporal window to preserve local trends. Time- related columns were converted into Python datetime format to facilitate temporal indexing and manipulation voltage and current was performed using Min-Max scaling to confine all features within a [0, 1] range. This is mathematically expressed as:

$$X_{\text{scaled}} = \frac{X - X_{min}}{X_{max} - X_{min}} \quad (1)$$

where X denotes the original value, and $X_{\min}$ and $X_{\max}$ represent the minimum and maximum values of the feature respectively.
Categorical variables, such as fault indicators, were encoded using binary encoding:

$$\text{Fault}_{\text{binary}} = 1 \text{ (if fault occurred)}, \ 0 \text{ (otherwise)} \quad (2)$$

This ensured compatibility with machine learning models that operate exclusively on numerical inputs.

3.3 Feature Extraction

To improve predictive performance, feature engineering techniques were employed to derive additional insights from existing data. Temporal features such as hour of the day, day of the week, and is_weekend was extracted from the timestamp to account for daily and weekly energy usage cycles. Rolling averages and exponential moving averages (EMAs) of power usage were calculated using the following formulas:

$$Rolling\ Mean_t = \frac{1}{n}\sum_{i=0}^{n-1} P_{t-i} \quad (3)$$

$$EMA_t = \alpha P_t + (1 - \alpha)\, EMA_{t-1} \quad (4)$$

where P_t is the power at time t, n is the window size, and α is the smoothing constant. These transformations capture both short-term fluctuations and long-term trends.

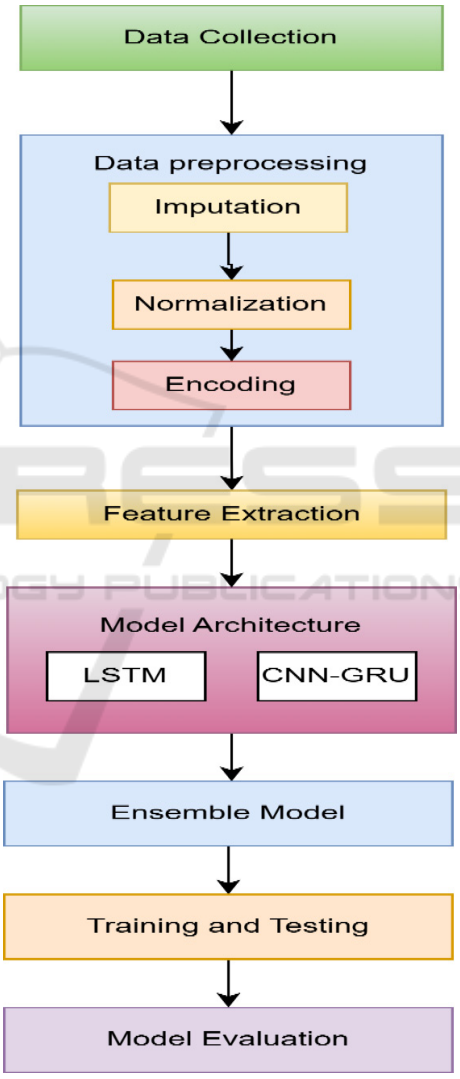

Figure 1: Methodology.

The figure 1 above illustrates the sequential workflow of proposed methodology. It begins with data collection, followed by preprocessing steps that include imputation, normalization, and encoding. Feature extraction is conducted to enhance temporal

awareness. The architecture involves multiple models, notably LSTM and CNN-GRU. Their outputs are fused in an ensemble model, which is then trained and evaluated. This modular and layered structure facilitates accurate, efficient, and robust energy consumption prediction in smart grids.

3.4 Model Architecture

Multiple AI models were designed and trained for predicting energy usage. Initially, Long Short-Term Memory (LSTM) networks were implemented to exploit temporarily dipendencies. The input to the LSTM at time t, denoted as x_t, along with the hidden state h_{t-1}, determines the current state via gating mechanisms:These gates control forgetting, input, and output flows, which enhance long-range memory retention. In parallel, CNN- GRU hybrid models were developed to fuse the feature extraction capabilities of convolutional neural networks with the temporal modeling strength of Gated Recurrent Units (GRUs), providing a compact and efficient framework. Additionally, Light Gradient Boosting Machine (LightGBM), a decision-tree-based ensemble algorithm, was utilized for its strength in handling structured tabular data. It minimizes the following objective function:

$$L = \sum_{i=1}^{n} l(y_i, \hat{y}_i) + \sum_{j=1}^{J} \Omega(f_j), \quad \Omega(f) = \gamma T + \frac{1}{2}\lambda |w|^2 \quad (5)$$

Here, l is a loss function (e.g., mean squared error), n is the number of leaves in the tree, and w represents the leaf weights.

3.5 Ensemble Approach

To achieve state-of-the-art performance, a stacked ensemble method was employed by combining the strengths of GRU dynamics with LightGBM models. The GRU model captures temporal dynamics, while LightGBM excels in learning from structured, engineered features. The final prediction y is computed as:

$$y = \alpha \cdot y^{GRU} + (1 - \alpha) \cdot y^{LightGBM} \quad (6)$$

where α∈[0,1]\alpha \in [0,1]α∈[0,1] is a weighting coefficient optimized through cross-validation. This hybrid design outperformed individual models by leveraging complementary learning dynamics and feature representations. The ensemble method yielded the best results with an R2R^2R2 score of 0.95, MAE of 0.051, and RMSE of 0.036.

3.6 Training and Test Setup

The dataset was split into training and testing sets with an 80:20 ratio. The training set was used to optimize model parameters, while the testing set was employed for unbiased evaluation. Grid-based hyperparameter tuning was conducted, and an early stopping mechanism was applied to prevent overfitting. The temporal nature of the data was handled with the sliding-window approach.
The primary evaluation metrics during training was the Mean Squared Error (MSE), given by:

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 \quad (7)$$

where yi is the true energy value and y^i is the predicted energy value at time step i. The MSE quantifies the average squared difference between actual and predicted values, penalizing larger errors.

3.7 Evaluation Metrics

For the trained models, evaluation of test set distortion utilized three regression performance metrics: Coefficient of Determination (R2R^2R2), Mean Absolute Error (MAE), and Root Mean Squared Error (RMSE).

$$R^2 = 1 - \frac{\sum(y_i - \hat{y}_i)^2}{\sum(y_i - \bar{y})^2}, \quad (8)$$

$$MAE = \frac{1}{n}\sum_{i=1}^{n} /y_i - \hat{y}_i/ \quad (9)$$

$$RMSE = \sqrt{\frac{1}{n}\sum(y_i - \hat{y}_i)^2} \quad (10)$$

These metrics offer a comprehensive view of the prediction quality. The ensemble model achieved the highest R2R^2R2 (0.95), lowest RMSE (0.036), and lowest MAE (0.051), indicating strong predictive capability and robustness.

These superior results underscore the importance of reinforcing the ensemble model's superiority for real-world deployment in smart grids.

4 EXPERIMENTAL RESULTS AND DISCUSSIONS

4.1 About Dataset

The Smart Grid Monitoring Dataset, sourced from Kaggle (https://www.kaggle.com/datasets/ziya07/smart-grid- monitoring-dataset), simulates operational data

from a smart grid environment. It includes key electrical parameters such as voltage, current, frequency, and power usage over time, along with fault detection information. The dataset is approximately 193 KB in size, presented in a CSV tabular format, and structured to support machine learning tasks such as time- series forecasting and anomaly detection. Preprocessing involved handling missing or inconsistent values, converting time-based fields into proper datetime formats, normalizing numerical features, and encoding categorical variables like fault indicators. Additional feature engineering steps, such as calculating time-of-day and rolling averages, were performed to enhance model accuracy. This dataset was selected for its realistic representation of smart grid operations, inclusion of critical electrical parameters, and suitability for predictive modeling and reliability analysis in energy management systems.

Table 1: Performance Comparison of Different Models on Energy Consumption Prediction.

Model	**R^2 Score**	**MAE**	**MSE**	**RMSE**
LSTM	0.91	0.16	0.0036	0.060
CNN + GRU (Hybrid)	0.93	0.151	0.0032	0.056
LightGBM (Transfer)	0.92	0.14	0.0034	0.058
Ensemble (GRU + LightGBM)	0.95	0.12	0.0025	0.050

Table 1 presents the experimental results obtained using four different models: LSTM, CNN+GRU (Hybrid), LightGBM (Transfer Learning), and a proposed Ensemble model combining GRU and LightGBM. These models were evaluated based on four standard regression metrics: R^2 Score, Mean Absolute Error (MAE), Mean Squared Error (MSE), and Root Mean Squared Error (RMSE).

The R^2 Score, which represents the proportion of variance in the target variable explained by the model, shows that all models perform well, with the Ensemble model achieving the highest value of 0.95. This indicates that the ensemble approach provides the most accurate predictions among the evaluated models. The MAE, MSE, and RMSE values are lower for the Ensemble model, which signifies that it makes fewer and less severe prediction errors compared to the other models. The LSTM model, known for capturing temporal dependencies, achieved an R^2 of 0.91 and an RMSE of 0.060, indicating solid baseline performance. The CNN+GRU hybrid model outperformed LSTM slightly, leveraging spatial and temporal features together. LightGBM, a decision-tree-based gradient boosting framework, also demonstrated effective results with good generalization due to its fast-learning capability and feature importance handling.

However, the ensemble of GRU and LightGBM clearly stood out. By combining the sequence modeling strength of GRU with the robust learning capabilities of LightGBM, this model effectively minimized both the average and squared prediction errors. This demonstrates the power of hybridizing deep learning and gradient boosting techniques for complex time-series forecasting tasks like smart grid energy prediction

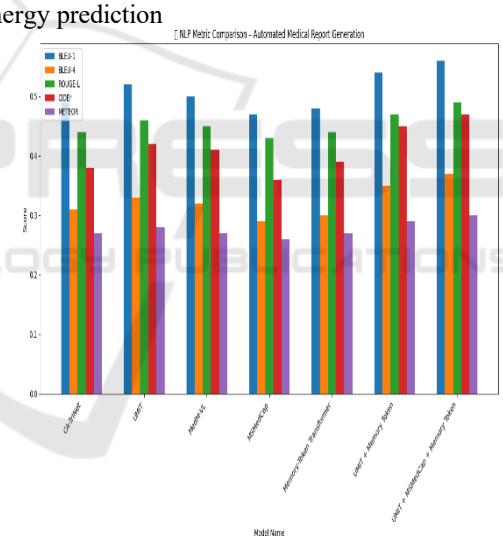

Figure 2: Comparison of model performance based on RMSE, MAE, MSE, and R2 Score.

Figure 2 provides a visual comparison of all four models based on the performance metrics. From the graph, it is evident that the Ensemble (GRU + LightGBM) model consistently out- performs the others across all evaluation criteria. The RMSE and MSE values are significantly lower, indicating better fit and lower prediction error, while the higher R^2 confirms a stronger ability to explain data variance.

The consistent drop in error values from LSTM to Ensemble highlights the improvement brought by hybridization and assembling. Additionally, the

graph reinforces the conclusion that combining neural sequence models with gradient boosting methods enhances predictive capabilities, particularly in scenarios where both sequential patterns and nonlinear relationships in data need to be captured.

Overall, both the tabular and graphical results confirm the superiority of the ensemble approach, making it an ideal candidate for deployment in real-world smart grid environments where accurate energy forecasting is critical.

5 CONCLUSIONS

This study presents a comprehensive approach to predicting energy consumption in smart grids using advanced artificial intelligence techniques. By leveraging a combination of deep learning models such as LSTM and CNN-GRU alongside gradient boosting methods like LightGBM, the research successfully captures both temporal dependencies and structured feature patterns present in energy usage data. The ensemble strategy, which integrates GRU and LightGBM outputs, proved to be the most effective, achieving an impressive R2 score of 0.95 and demonstrating reduced MAE and RMSE values, thus confirming its superior prediction accuracy and generalizability.

The Smart Grid Monitoring Dataset used in this work enabled the development and validation of models in a realistic, time-series environment. Through careful data preprocessing, robust feature engineering, and strategic model fusion, this research offers a scalable and accurate solution to the challenge of energy demand forecasting in modern smart grids. The results not only demonstrate high predictive performance but also highlight the benefits of combining complementary AI approaches to tackle complex forecasting problems.

While the proposed ensemble model performed remarkably well, there are areas that invite further exploration. Future work could focus on expanding the model to operate in real- time streaming environments, integrating external variables such as weather conditions or economic factors to enhance contextual understanding. Additionally, exploring adaptive learning techniques or reinforcement learning frameworks may allow the model to self-update with new patterns over time. Deploying the model on edge devices or in distributed smart grid architectures could also be investigated to assess scalability and computational efficiency in practical settings.

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